

Haebeom Jung

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Education

Seoul National University

PH.D. STUDENT IN ARTIFICIAL INTELLIGENCE

- **Current GPA:** 4.04/4.30
- A member of the SNU VGI Lab (Advisor: Prof. Jaesik Park)
- Received full scholarship for 5 semesters.

Seoul, South Korea

Sep. 2023 - Present

KwangWoon University

B.S. IN DIVISION OF ROBOTICS

- **GPA:** 3.81/4.50, **Major GPA:** 3.93/4.50
- Received full scholarship for 6 semesters.

Seoul, South Korea

Mar. 2016 - Feb. 2023

Research in Progress

History-Aware Vision-Language Models for Robotic Grasping

HAEBEOM JUNG*, **HYUNJOON LEE***, **DAE-UN LEE**, **JAESUNG CHOI**, **JAESIK PARK**

2026 - Present

- VLM-based grasping is typically evaluated from a single static observation, overlooking temporal cues available during interaction. We study both static-view and video-based grasping, using interaction history when temporal context is available.

Blind-in-Hand: Vision-Free Manipulation under Uncertain Grasps

JOONGON CHOI, **HAEBEOM JUNG**, **JAESIK PARK**

2026 - Present

- Robots often lose visual access during contact-rich in-hand manipulation, making recovery from unstable grasps difficult. Blind-in-Hand develops vision-free reinforcement learning policies that use force and proprioception to manipulate unseen objects robustly.

Open-X-Tactile: OMY-Based Tactile Data Contribution

DATA CONTRIBUTOR

2026 - Present

- Generalist tactile policies require diverse data across sensors and robot embodiments. We contribute tactile manipulation data collected with a tactile-equipped OMY robot to expand the cross-platform coverage of Open-X-Tactile.

DROID-Flow: Toward SLAM in the Wild

NAMTAE KIM, **HAEBEOM JUNG**, **JAESIK PARK**

2026 - Present

- Real-world SLAM remains brittle when optical-flow correspondences fail to respect object parts in dynamic, unconstrained scenes. DROID-Flow addresses this limitation through multi-view flow reasoning and uncertainty-aware estimation.

Publications

Distractor-Aware COLMAP-Free 3D Gaussian Splatting

Under Review

NAMTAE KIM, **HAEBEOM JUNG**, AND **JAESIK PARK**

Articulated Object Reconstruction from Rest-State Observation

ECCV 2026

DAEUN LEE, **JAEAH LEE**, **WOOSUNG KIM**, **HAEBEOM JUNG**, AND **JAESIK PARK**

Sep. 2026

Targetless LiDAR-Camera Calibration with Neural Gaussian Splatting

RA-L 2026

HAEBEOM JUNG, **NAMTAE KIM**, **JUNGWOO KIM**, AND **JAESIK PARK**

Apr. 2026

Metropolis-Hastings Sampling for 3D Gaussian Reconstruction

NeurIPS 2025

HYUNJIN KIM, **HAEBEOM JUNG**, AND **JAESIK PARK**

Dec. 2025

Professional Experience

Kakao Mobility

SOFTWARE · ROBOTICS ENGINEER (FULL-TIME)

- Developed a data collection system for autonomous vehicles.
- Researched visual SLAM and localization for real-world driving environments.

Pangyo, South Korea

Dec. 2021 - Apr. 2022

Stryx

SOFTWARE · ROBOTICS ENGINEER (COMPULSORY MILITARY SERVICE)

- Led software development for the hardware business unit.
- Developed a sensor data collection system for a mobile mapping platform.
- Researched and developed autonomous navigation for unmanned ground vehicles, including SLAM, localization, sensor fusion, and path planning.

Seoul, South Korea

Dec. 2019 - Dec. 2021

ROBOTIS

ROBOTICS ENGINEER (INTERN)

Seoul, South Korea

Jan. 2019 - Feb. 2019

- Validated ROBOTIS OpenManipulator prior to its open-source release.
- Developed and published an example application for ROBOTIS OpenManipulator SARA.

Skills

Programming	Python, C/C++, Java
Robotics & Tools	PyTorch, ROS/ROS 2, Isaac Sim, Qt
Languages	Korean (native), English (fluent)

Open Source Contributions

Selected Robotics Projects

- Maintainer of **ORB-SLAM3-ROS2**, a ROS 2 integration for the ORB-SLAM3 library.
- Contributor to **FAST-LIO**, fixing Velodyne LiDAR integration issues.
- Contributor to **LIO-SAM**, adding a ROS service for map saving.

Patents

Device and Method for Targetless LiDAR-Camera Calibration

Seoul National Univ., South Korea

HAEBEOM JUNG, NAMTAE KIM, AND JAESIK PARK

May. 2025

Application No. 10-2025-0067792

Method of estimating the location of a moving object using vector map

Kakao Mobility, South Korea

YESUNG HWANG, HAEBEOM JUNG, JEONGMIN KANG, ILSUK PARK, SEUNGKWAN SONG, AND SEUNGHWAN HONG

Jan. 2024

Registration No. 10-2624-6440000

Awards

ROBOTICS COMPETITIONS

INTERNATIONAL

2018	3rd Place , ROBOCUP CANADA, KidSize Humanoid Technical Challenge [Video]	Montreal, Canada
2017	4th Place , ROBOCUP IRAN, KidSize Humanoid Soccer [Video]	Tehran, Iran

DOMESTIC

2018	3rd Place , R-BIZ Challenge, Turtlebot3 Autorace [Video]	Ilsan, South Korea
2018	2nd Place , IRC, Intelligence Humanoid Robot Sports	Ilsan, South Korea
2017	Winner , IRC, Intelligence Humanoid Robot Sports [Video]	Ilsan, South Korea
2017	Winner , ROBOCUP KOREA, Intelligence Humanoid Soccer	Daegu, South Korea
2016	3rd Award , KIRO, Future-Robot Idea Presentation	Pohang, South Korea

HONORS

2019	Outstanding Mentor , 2019 Samsung Electronics Junior Software Creation Contest	Samsung Electronics
2017	Achievement Award , Spread the name of KwangWoon University throughout the year	KwangWoon Univ.

SCHOLARSHIPS

2026	Full scholarship , Jeongheon Foundation Scholar	Jeongheon Foundation
2023-2025	Full scholarship , AI Fellowship	Seoul National Univ.
2017-2019	Full scholarship , Robotics Competition Team	KwangWoon Univ.

Fellowships and Research Funding

Photo-Realistic Rendering for Surrounding Dynamic Mobile Scenes

Ministry of Education, South Korea

MASTER'S RESEARCH FELLOWSHIP PROGRAM

Jul. 2024 - Jun. 2025

Research Service & Robotics Activities

Invited Talks

NAVER LABS, Apr. 2025

Reviewer

- CVPR (2026)
- ECCV (2026)
- NeurIPS (2026)
- IEEE TIP (2025)

Kakao Mobility NEMO Conference

AUTONOMOUS DRIVING ROBOT DOCENT

Seoul, South Korea

Feb. 2022

Explained and operated an autonomous robot that travels inside the conference hall. **[Video]**

Samsung Electronics Junior Software Creation Contest

MENTOR

- Serviced the app, which provided a system for managing kindergarten students going to and from school and attendance using beacons.
- The high school team I coached won the grand prize among 10,000 teams.

Suwon, South Korea

Jul. 2019 - Sep. 2019

RO:BIT (Robot Game Sports Team)

MEMBER & TEAM LEADER AT 2018

- As a member of the humanoid team for 3 years, I participated in many external activities and competitions related to robots.
- I mainly researched and developed a robot algorithm called humanoid intelligence in games such as soccer, basketball and athletics, etc.
- In 2018, I led the team as the leader of the humanoid team and won awards in international competitions.

Seoul, South Korea

Jul. 2016 - Dec. 2018